

DELETING THE POWERS OF FOUR: STRICTLY INCREASING REPRESENTATION FUNCTIONS

VAMSHI JANDHYALA

ABSTRACT. For $A \subseteq \mathbb{N} = \{0, 1, 2, \dots\}$ let $R_{A,h}(n)$ count the ordered h -tuples from A with sum n . Sándor and Yang asked whether $R_{A,h}$ is strictly increasing when $A = \mathbb{N} \setminus \{4^{j+2} : j \geq 0\}$. It is, for every $h \geq 3$ and from $n = 0$ on. Bell and Shallit's Theorem 1 already gives strict increase for all large n ; the point here is the explicit bound $R_{A,3}(N) - R_{A,3}(N-1) \geq N+1-3m-m^2$, with m the number of deleted elements up to N , which holds for every deleted set and is positive at every N for the powers of four. The argument is the coefficient computation of Bell and Shallit's Theorem 2. We also record two facts about Sándor and Yang's Problem 1.4, which asks for densities $d < 1$: no eventually periodic set of density strictly between 0 and 1 works, and any set that works has $A(N) \geq ((N+1)(N+2)/2)^{1/3}$. The bound, the reduction to $h = 3$ and Proposition 1 are formally verified in Lean 4.

1. THE QUESTION

Write $R_{A,h}(n)$ for the number of ordered h -tuples $(a_1, \dots, a_h) \in A^h$ with $a_1 + \dots + a_h = n$. For $A = \mathbb{N}$ every tuple counts and $R_{\mathbb{N},3}(n) = \binom{n+2}{2}$, which increases. Delete a few elements and some tuples disappear. Can $R_{A,3}$ still increase strictly if infinitely many elements are deleted?

For two summands it cannot: by a theorem of Erdős, Sárközy and Sós, $R_{A,2}$ is eventually increasing only when A is cofinite [3]. For three or more summands it can. Bell and Shallit [1] proved that deleting any sufficiently sparse set F leaves $R_{\mathbb{N} \setminus F, h}$ eventually strictly increasing for $h \geq 3$, and showed that deleting $\{2^{j+2} - 1 : j \geq 0\} = \{3, 7, 15, \dots\}$ gives strict increase from $n = 0$. Sándor and Yang [3] gave the example $\mathbb{N} \setminus \{2^j : j \geq 10\}$ and wrote (Remark 1.3): *it would be interesting to determine whether the set $\mathbb{N} \setminus \{4^{n+2} : n \geq 0\}$ from Bell and Shallit also has this property.*

Try it before reading on. The first deleted number is 16. Up to $N = 15$ nothing is missing and $R_{A,3}(N) = \binom{N+2}{2}$. At $N = 16$ the three triples $(16, 0, 0)$, $(0, 16, 0)$, $(0, 0, 16)$ disappear, so $R_{A,3}(16) = 153 - 3 = 150$, still above $R_{A,3}(15) = 136$. The loss grows with each new power of four, but so does N . Figure 1 shows the race up to 300: the loss stays in single digits.

Proposition 1. *Let $A = \mathbb{N} \setminus \{4^{j+2} : j \geq 0\}$. For every $h \geq 3$ and every $n \geq 0$, $R_{A,h}(n+1) > R_{A,h}(n)$.*

Two remarks on sources. The set $\{4^{j+2}\}$ does not appear in the arXiv version of Bell and Shallit [1], whose explicit example is $\{2^{j+2} - 1\}$, already settled there; so Remark 1.3 may have meant that set. Either way, Proposition 1 answers the remark as printed. Second, in the text layer of the arXiv PDF of [3] the exponent is flattened, and $\{4^{n+2}\}$ reads as $\{4n+2\}$. That progression gives a different answer, and it is the example of Section 3.

2. THE COEFFICIENT BOUND

Let $E = \mathbb{N} \setminus A$ be the deleted set, and write $m(N) = \#(E \cap [0, N])$ and $r_{E,k}(N)$ for the number of ordered k -tuples from E with sum N , with $r_{E,k}(-1) = 0$. In terms of generating functions, with $D(x) = \sum_{e \in E} x^e$ and $G(x) = \sum_{a \in A} x^a = (1-x)^{-1} - D(x)$, we have $\sum_n R_{A,h}(n)x^n = G(x)^h$, so the increments of $R_{A,3}$ are the coefficients of $(1-x)G(x)^3$. Expanding the cube and using $(1-x) \cdot (1-x)^{-1} = 1$,

$$(1) \quad (1-x)G(x)^3 = (1-x)^{-2} - 3D(x)(1-x)^{-1} + 3D(x)^2 - (1-x)D(x)^3.$$

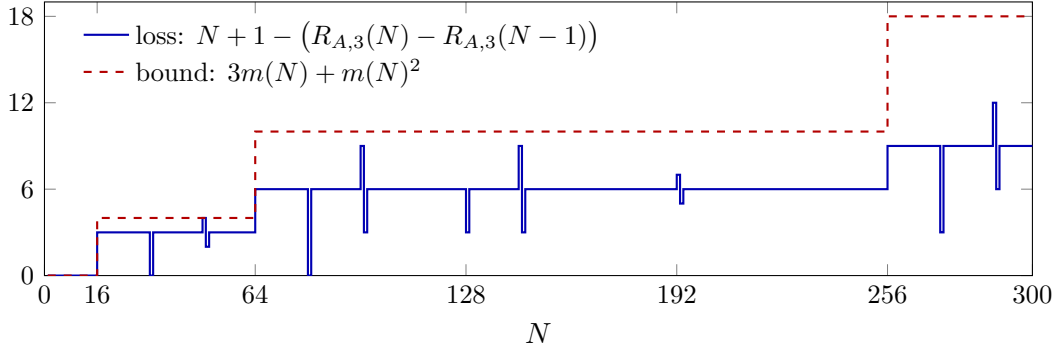


Figure 1. For $A = \mathbb{N} \setminus \{16, 64, 256, \dots\}$, how much smaller the increment of $R_{A,3}$ is than the increment $N+1$ for $A = \mathbb{N}$ (solid), and the upper bound $3m+m^2$ from Lemma 2 (dashed). The loss is at most 12 up to $N = 300$ while $N+1$ grows, so every increment is positive; the dips below the bound come from the terms $3r_{E,2}(N)$ and $r_{E,3}(N-1)$ at sums of powers of four.

Read each term at x^N . The first is $N+1$, the increment of $\binom{N+2}{2}$, as if nothing were deleted. The second is $3m(N)$: one unit lost for each deleted element up to N , in each of the three positions. The third, $3r_{E,2}(N)$, adds back what was subtracted twice. The last is $r_{E,3}(N) - r_{E,3}(N-1)$.

Lemma 2. For every $E \subseteq \mathbb{N}$ and every $N \geq 1$,

$$R_{A,3}(N) - R_{A,3}(N-1) \geq N+1 - 3m(N) - m(N)^2.$$

Proof. By (1), the increment equals $N+1 - 3m(N) + 3r_{E,2}(N) - r_{E,3}(N) + r_{E,3}(N-1)$. Drop the two nonnegative terms $3r_{E,2}(N)$ and $r_{E,3}(N-1)$. A triple from E with sum N is fixed by its first two entries, both in $E \cap [0, N]$, so $r_{E,3}(N) \leq m(N)^2$. $\square$

The case $h = 3$ carries the rest. The coefficients of $(1-x)G^h$ are the increments of $R_{A,h}$, and

$$(1-x)G(x)^h = ((1-x)G(x)^3)G(x)^{h-3}.$$

Lemma 3. If $0 \in A$ and $R_{A,3}(n+1) > R_{A,3}(n)$ for all $n \geq 0$, then $R_{A,h}(n+1) > R_{A,h}(n)$ for all $h \geq 3$ and $n \geq 0$.

Proof. Every coefficient of $F = (1-x)G^3$ is positive: the one at x^0 is $R_{A,3}(0) = 1$ because $0 \in A$, the others are the increments. The series G^{h-3} has nonnegative coefficients and constant term 1. So the coefficient of x^{n+1} in FG^{h-3} is at least $[x^{n+1}]F \cdot 1 > 0$. $\square$

Proof of Proposition 1. By Lemma 3 it suffices to take $h = 3$. If $m = m(N) = 0$ the bound of Lemma 2 is $N+1 > 0$. If $m \geq 1$, the deleted elements up to N are $4^2, \dots, 4^{m+1}$, so $N \geq 4^{m+1}$ and the bound is at least $4^{m+1} + 1 - 3m - m^2$. This is 13 at $m = 1$, and going from m to $m+1$ adds $3 \cdot 4^{m+1} - 2m - 4 > 0$. $\square$

Nothing here is specific to powers of four. Lemma 2 gives strict increase from 0 for any deleted set with $0 \notin E$ and $3m(N) + m(N)^2 \leq N$ for all N . Deleting the powers of two from 2^{10} on, Sándor and Yang's own example, also satisfies this.

3. TWO FACTS ABOUT PROBLEM 1.4

The sets above have density 1. Sándor and Yang ask (Problem 1.4) for which densities $d < 1$ there is a set A with $A(n)/n \rightarrow d$ and every $R_{A,h}$, $h \geq 3$, strictly increasing. By Lemma 3 this is a question about $R_{A,3}$ alone, together with $0 \in A$.

The first candidates to try are periodic. Take $A = \{n : n \not\equiv 2 \pmod{4}\}$, of density $3/4$. Then $R_{A,3}$ begins 1, 3, 3, 4, 9, 12, 12, 15, 24, 28, 27, $\dots$: it stalls at $n = 2$ and $n = 6$, and from $n = 10$ on it drops at every $n \equiv 2 \pmod{4}$. This is general.

Proposition 4. Let $A \subseteq \mathbb{N}$ agree, outside a finite set, with a periodic set of density strictly between 0 and 1. Then for every $h \geq 3$ there are infinitely many n with $R_{A,h}(n+1) < R_{A,h}(n)$.

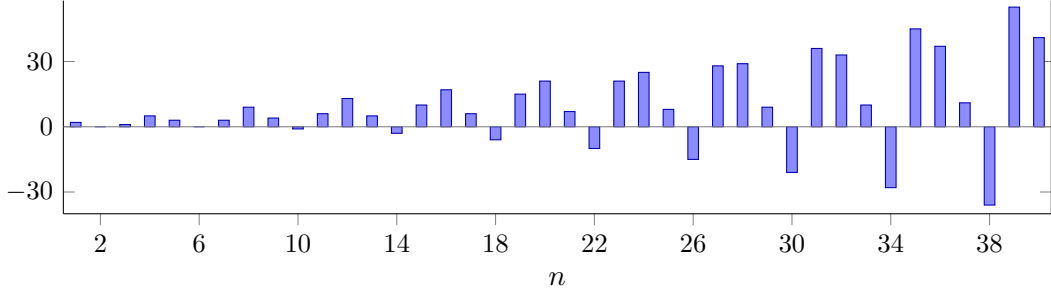


Figure 2. Increments $R_{A,3}(n) - R_{A,3}(n-1)$ for the periodic set $A = \{n : n \not\equiv 2 \pmod{4}\}$. The increments differ from class to class, and on the class $n \equiv 2 \pmod{4}$ (the labelled ticks) they are zero or negative and fall quadratically, like a negative multiple of n^2 , as the proof of Proposition 4 predicts.

Proof. Let q be the period and $a_r \in \{0, 1\}$ ($0 \leq r < q$) the indicator of residue r in the periodic set. Let c_s be the number of h -tuples $(r_1, \dots, r_h)$ of allowed residues with $r_1 + \dots + r_h \equiv s \pmod{q}$. A fixed residue tuple with integer sum T contributes $\binom{(n-T)/q+h-1}{h-1}$ representations of each $n \equiv T$, and the finitely many changes affect only $O(n^{h-2})$ representations. Hence

$$R_{A,h}(n) = \frac{c_{n \bmod q}}{(h-1)!q^{h-1}} n^{h-1} + O(n^{h-2}).$$

The sequence (c_s) is the h -fold cyclic convolution of (a_r) , so its discrete Fourier transform at a q -th root of unity ζ is $\hat{a}(\zeta)^h$. If (c_s) were constant then $\hat{a}(\zeta) = 0$ for every $\zeta \neq 1$, and (a_r) would be constant, giving density 0 or 1. So (c_s) is not constant, and going once around the cycle there is an s with $c_{s+1} < c_s$ (indices mod q). For $n \equiv s$ the leading term of $R_{A,h}(n+1) - R_{A,h}(n)$ is $(c_{s+1} - c_s)n^{h-1}/((h-1)!q^{h-1}) < 0$, so the increment is negative for all large n in that class. $\square$

So a set answering Problem 1.4 with $0 < d < 1$ must be aperiodic, like Dombi's density-1/2 set, which Sándor and Yang found fails strictly at $n = 23$ [3]. How sparse can it be?

Proposition 5. *If $R_{A,3}$ is strictly increasing from $n = 0$, then $A(N)^3 \geq (N+1)(N+2)/2$ for every N .*

Proof. Since $R_{A,3}(1) > R_{A,3}(0) \geq 0$, the number 1 has a representation, so $0 \in A$ and $R_{A,3}(0) = 1$. Strict increase then gives $R_{A,3}(n) \geq n+1$. Every triple with sum at most N has its entries in $A \cap [0, N]$, so $A(N)^3 \geq \sum_{n \leq N} R_{A,3}(n) \geq \sum_{n \leq N} (n+1)$. $\square$

So $A(N) \gtrsim N^{2/3}$, which still allows density 0; Problem 1.4 at $d = 0$ is not settled by this.

Remark 6. Aperiodic candidates are easy to write down and hard to prove. One example: delete the blocks $[(4k)^2, (4k+1)^2 - 1]$ for $k \geq 2$. The resulting set has density $3/4$ (the k -th block has length $8k+1$ and starts at $16k^2$), and an exact computation shows $R_{A,3}$ strictly increasing for $0 \leq n < 262,144$. We have no proof that it stays increasing, so this is evidence, not a result.

4. VERIFICATION

Lemmas 2 and 3 and Proposition 1 are proved in Lean 4 with Mathlib [2], using only the standard axioms. There $R_{A,h}(N)$ is the coefficient of x^N in G^h , identity (1) is a ring identity in the power series ring $\mathbb{Z}[[x]]$, and for $h = 3$ the coefficient is shown to equal the number of ordered triples. Propositions 4 and 5 are proved by hand only. Two programs check the statements numerically: one computes $R_{A,3}$, $R_{A,4}$ and $R_{A,5}$ by direct convolution up to $N = 600$ and checks the bound of Lemma 2 at every N ; the other checks the identity behind the bound through $N = 10^6$ using only the deleted set. Code and Lean files are at [4].

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LONDON, UNITED KINGDOM

Email address: `vamshi.jandhyala@gmail.com`